7.2.1 Banded matrices

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Week 7-8 due Mar 2, 2025 01:52 IST

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actually symmetric

positive-definite. And here we're just going to for the moment give up on

complex values. Let's just stay with real numbers. Now, what did we learn about

factoring symmetric positive-definite matrices? Well, we learned that we can

deploy the Cholesky factorization for that. And how would that algorithm

work? Well that algorithm would partition our matrix and would then take the

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2.0x

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Reading assignment

1/1 point (graded)

Read Unit 7.2.1 of the notes. [LINK]

- [x] done

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Bandwidth of a diagonal matrix
This just came up in course discussion, so wanted to check: In https://www....
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? Homework 7.2.1.1 Solution requires faith?
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Homework 7.2.1.1

1/1 point (graded)

The 1D equivalent of the example from Unit 7.1.1.1 is given by the tridiagonal linear system

$$A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

Prove that this linear system is nonsingular.

(This is a pretty tricky problem. Feel free to sneak a peak at the solution.)

☑ done

✔

Solution:

For a complete solution see the notes.

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Answers are displayed within the problem

Homework 7.2.1.2

0.9375/1 point (graded)

Compute the Cholesky factorization of

$$A = \begin{pmatrix} 4 & -2 & 0 & 0 \\ -2 & 5 & -2 & 0 \\ 0 & -2 & 10 & 6 \\ 0 & 0 & 6 & 5 \end{pmatrix}.$$

2	1	0
✔ Answer: 2	✖ Answer: 0	✔ Ar
-1	2	0
✔ Answer: -1	✔ Answer: 2	✔ Ar
0	-1	3
✔ Answer: 0	✔ Answer: -1	✔ Ar
0	0	2
✔ Answer: 0	✔ Answer: 0	✔ Ar

Solution:

For a complete solution see the notes.

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Homework 7.2.1.3

1/1 point (graded)

Let $A \in \mathbb{R}^{n \times n}$ be tridiagonal and SPD so that

$$A = \begin{pmatrix} \alpha_{0,0} & \alpha_{1,0} & & & \\ \alpha_{1,0} & \alpha_{1,1} & \alpha_{2,1} & & \\ & \ddots & \ddots & \ddots & \\ & & \alpha_{n-2,n-3} & \alpha_{n-2,n-2} & \alpha_{n-1,n-2} \\ & & & \alpha_{n-1,n-2} & \alpha_{n-1,n-1} \end{pmatrix}.$$

- Propose a Cholesky factorization algorithm that exploits the structure of this matrix.

(selected) done

- What is the cost? (Count square roots, divides, multiplies, and subtractions.)

( ) n square roots, $n-1$ divides, $\frac{1}{3}n^3$ multiplies, and $\frac{1}{3}n^3$ subtracts.

( ) $2n$ square roots, $2n-1$ divides, $2n-1$ multiplies, and $2n-1$ subtracts.

(selected) n square roots, $n-1$ divides, $n-1$ multiplies, and $n-1$ subtracts.

- What would have been the (approximate) cost if we had not taken advantage of the tridiagonal structure?

( ) n^3

(selected) $\frac{1}{3}n^3$

( ) $\frac{2}{3}n^3$

Solution:

For a complete solution see the notes.

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Homework 7.2.1.4

1/1 point (graded)
Propose an algorithm for overwriting y with the solution to A x = y for the SPD matrix in Homework 7.2.1.3.

- [x] done

Solution:

For a complete solution see the notes.

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let's get away from the specific example and look at the general symmetric positive-definite matrix that has some kind of banded structure. So here we have a matrix where each x indicates some value that may or may not be 0, but all other values are equal to 0. And this particular matrix has a band width of 3. So the number of diagonals and subdiagonals that have nonzeros in them is the bandwidth. In

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Homework 7.2.1.5

1/1 point (graded)
Assume the SPD matrix $A \in \mathbb{R}^{m \times m}$ has a bandwidth of b. Propose a modification of the right-looking Cholesky factorization

$A = \text{Chol-right-looking}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$

A_{TL} is 0×0

while $n(A_{TL}) < n(A)$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array}\right)$$

$\alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}}$

$a_{21} := l_{21} = a_{21}/\alpha_{11}$

$A_{22} := A_{22} - a_{21}a_{12}^T$ (syr: update only lower triangular part)

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right)$$

endwhile

that takes advantage of the zeroes in the matrix. (You will want to draw yourself a picture.) What is its approximate cost in flops (when m is large)?

- (●) Approximately nb^2 flops
- ( ) Approximately $\frac{1}{3}n^3$ flops
- ( ) Approximately $\frac{1}{3}nb^2$ flops

✔

Solution:

For a complete solution see the notes.

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here is b- 1 so it would have been 2 times b minus 1 squared.

If we take advantage of symmetry, that's approximately divided by 2. And roughly speaking we need

to do that for every entry on the diagonal.

Of course, once you run out of matrix here, there would be less work to be done. But if

the matrix is large enough, that is roughly what would happen. So we would have to do this -- let's

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